

3 (**short**) Fig. 3 shows an AC circuit connected to a 240 V, 50 Hz power supply. Determine the total complex impedance of the circuit and hence find the magnitude and phase of the input current with respect to the voltage source. [10]

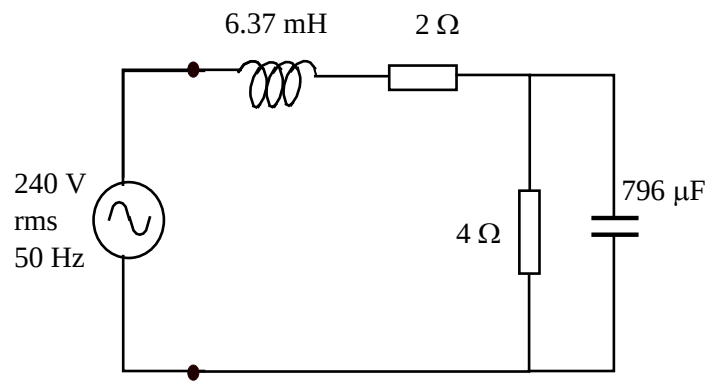


Fig. 3